Hospital Booking System Specification

Purpose

The purpose of this Hospital Booking System is to provide a convenient way for patients to book appointments, view doctor information and availability, check token numbers and queue status, and cancel or reschedule appointments online. It also helps administrators manage doctors, schedules, appointments, and patient queues efficiently, reducing manual work and improving the overall appointment booking process.

Functional Requirements Specification

FR-1: User Registration and Login

1.1 Description:

The system shall allow patients to create an account and log in using their registered credentials before accessing appointment and queue-related functions. The system shall associate each patient's appointments and queue information with the corresponding patient account.

1.2 Actor:
Patient

1.3 Precondition:

- The patient can access the system.
- The patient has not logged in yet.

1.4 Main Flow:

1. The patient opens the registration or login page.
2. For registration, the patient enters the required registration information.
3. The system validates the submitted information.
4. If the information is valid, the system creates a patient account.
5. For login, the patient enters their registered credentials.
6. The system verifies the credentials.
7. If the credentials are valid, the system grants access to the patient.

1.5 Alternative Flow:

- If the registration information is invalid, the system displays an error message and requests correction.
- If the login credentials are invalid, the system displays an error message and does not grant access.

1.6 Expected Result:
The patient can access the system using a valid account, and their appointment and queue information is associated with that account.

FR-2: View Doctor Information and Availability

2.1 Description:
The system shall allow patients to view doctors' information, specializations, and available appointment dates and times.

2.2 Actor:
Patient

2.3 Precondition:

- The patient can access the doctor information page.
- Doctor information and schedules are available in the system.

2.4 Main Flow:

1. The patient opens the doctor list.
2. The system displays the available doctors and their information.
3. The patient selects a doctor.
4. The system displays the selected doctor's details and specialization.
5. The system displays the available appointment dates and time slots.
6. The patient reviews the available schedules.

2.5 Alternative Flow:

- If no doctors are available, the system displays an appropriate message.
- If the selected doctor has no available appointment slots, the system informs the patient.
- The system prevents the patient from selecting unavailable time slots.

2.6 Expected Result:
The patient can view doctor information and identify available dates and times for an appointment.

FR-3: Book an Appointment

3.1 Description:
The system shall allow patients to book an appointment online by selecting a doctor, an available date, and an available time slot.

3.2 Actor:
Patient

3.3 Precondition:

- The patient is logged in.
- The selected doctor has available appointment slots.
- The patient can access the appointment booking function.

3.4 Main Flow:

1. The patient opens the appointment booking page.
2. The patient selects a doctor.
3. The system displays the doctor's available dates and times.
4. The patient selects an available date and time.
5. The system displays the selected appointment details.
6. The patient confirms the booking.
7. The system verifies that the selected time slot is still available.
8. The system creates and saves the appointment.
9. The system generates a token number for the appointment.
10. The system displays the booking confirmation and token number.

3.5 Alternative Flow:

- If the selected time slot is no longer available, the system prevents the booking and asks the patient to select another available slot.
- If the booking cannot be saved, the system displays an appropriate error message.

3.6 Expected Result:
The appointment is successfully recorded, and the patient receives a token number associated with the appointment.

FR-4: Check Current Token Queue

4.1 Description:
The system shall allow patients to view their token number, the current queue position, and their appointment status.

4.2 Actor:
Patient

4.3 Precondition:

- The patient is logged in.
- The patient has a valid appointment and an associated token number.

4.4 Main Flow:

1. The patient opens the queue status page.
2. The system retrieves the patient's appointment and token information.
3. The system retrieves the current queue information.
4. The system displays the patient's token number.
5. The system displays the current token being served and the patient's queue position.
6. The system displays the appointment or token status.
7. The system updates the displayed information when the queue status changes.

4.5 Alternative Flow:

- If the patient has no appointment or token, the system displays an appropriate message.
- If the queue information cannot be retrieved, the system informs the patient that the information is temporarily unavailable.

4.6 Expected Result:
The patient can view their token number, current queue information, and updated token status.

FR-5: Cancel or Reschedule an Appointment

5.1 Description:
The system shall allow patients to cancel or reschedule their upcoming appointments, subject to the appointment's current status and the availability of alternative time slots.

5.2 Actor:
Patient

5.3 Precondition:

- The patient is logged in.
- The patient has an upcoming appointment.

- The appointment belongs to the logged-in patient.

5.4 .1 Main Flow — Cancel Appointment:

1. The patient opens the appointment details page.
2. The system displays the patient's upcoming appointments.
3. The patient selects an appointment to cancel.
4. The system displays the selected appointment details.
5. The patient confirms the cancellation.
6. The system changes the appointment status to Cancelled.
7. The system updates the appointment and queue records as applicable.
8. The system displays a cancellation confirmation.

5.4.2 Main Flow — Reschedule Appointment:

1. The patient selects an upcoming appointment.
2. The patient chooses the reschedule option.
3. The system displays available alternative dates and time slots.
4. The patient selects a new available date and time.
5. The system verifies the availability of the selected slot.
6. The system updates the appointment with the new date and time.
7. The system displays the updated appointment details.

5.6 Alternative Flow:

- If the selected appointment is not eligible for cancellation or rescheduling, the system informs the patient.
- If the selected reschedule slot is unavailable, the system asks the patient to choose another slot.
- If the appointment update fails, the system displays an error message.

5.7 Expected Result:
The appointment is cancelled or updated to the newly selected date and time. The system reflects the changes in the appointment records and queue information as applicable.

FR-6: Manage Doctors, Schedules, Appointments, and Token Queue

6.1 Description:
The system shall allow administrators to manage doctor information, doctor schedules, patient appointments, and token queues.

6.2 Actor:
Admin

6.3 Precondition:

- The administrator is logged in.
- The administrator has permission to manage the relevant records.

6.4.1 Main Flow — Manage Doctors and Schedules:

1. The admin opens the management dashboard.
2. The system displays the available management functions.
3. The admin adds, edits, or removes doctor information.
4. The admin creates or updates doctor schedules.
5. The system validates and saves the changes.

6. The system reflects updated doctor information and availability in the patient interface.

6.4.2 Main Flow — Manage Appointments and Token Queue:

1. The admin opens the appointment management page.
2. The system displays appointment records and their current statuses.
3. The admin reviews appointments and updates their statuses when necessary.
4. The system displays the active token queue in the correct order.
5. The admin calls the next eligible waiting patient.
6. The system updates the called patient's token status.
7. The system removes cancelled tokens from the active waiting queue.
8. The system updates the queue information displayed to patients.

6.5 Alternative Flow:

- If the admin submits invalid information, the system displays an error message and requests correction.
- If a doctor or schedule cannot be updated, the system informs the admin.
- If no eligible patient is waiting, the system indicates that no patient is available to be called next.

6.6 Expected Result:

The administrator can manage doctor information, schedules, appointments, and token queues. Changes are saved and reflected in the relevant patient-facing information.